State-Space Models of Single, Double, and Triple Inverted Pendulums on a Cart

Abstract

This document derives continuous-time state-space models for one, two, and three serial inverted pendulums mounted on a translating cart. The derivation begins with the link kinematics and Lagrangian, linearizes the equations about the upright equilibrium, and then substitutes the numerical parameters of the physical systems. Cart friction and joint damping are defined as parameters for later use, but are set equal to zero in the present plant models.

Contents

1 Modeling assumptions

All three mechanisms move in the vertical x–z plane. The following assumptions define the mathematical model used in this derivation:

- links are rigid and joints have no clearance;
- the cart moves only horizontally;
- the input u is the horizontal force applied to the cart;
- the equilibrium angles are small enough for a first-order linearization about the upright position;
- friction and damping parameters are retained in the notation, but are set to zero when the state matrices are evaluated.

2 Parameters and coordinates

2.1 Parameter definitions

Symbol	Meaning
x	horizontal displacement of the cart
M	cart mass
n	number of links: $n \in \{1, 2, 3\}$
m_i	mass of link i
L_i	distance from joint i to joint $i+1$ (full length of link i)
ℓ_i	distance from joint i to the center of mass of link i
J_i	mass moment of inertia of link i about its center of mass and the hinge axis
θ_i	absolute angle of link i, measured from the upward vertical
g	gravitational acceleration
u	horizontal force applied to the cart
b_c	optional viscous damping coefficient of the cart
b_i	optional viscous damping coefficient of hinge i
$f_c,\ f_i$	optional dry-friction force/torque magnitudes

The undamped, frictionless model derived here uses

$$b_c = b_1 = b_2 = b_3 = 0, \qquad f_c = f_1 = f_2 = f_3 = 0. \tag{1}$$

These symbols are defined now so that damping or friction can be added later without changing the coordinate definitions.

2.2 Numerical values

The following numerical values are used in the MuJoCo model. Each pole has dimensions $0.04 \times 0.02 \times 0.30$ m. Its inertia about the hinge-axis direction through its center is

$$J_i = \frac{m_i}{12}\left(0.04^2 + 0.30^2\right) = 0.00381667 \text{ kg m}^2. \tag{2}$$

Parameter	Description	SI value used here
M	cart mass	2.0 kg
m_i	each pole mass	0.5 kg
L_i	pole length	0.30 m
ℓ_i	center-of-mass offset	0.15 m
J_i	from (2)	0.00381667 $\mathrm{kg\,m^2}$
g	gravity magnitude	9.81 $\mathrm{m/s^2}$

3 Kinematics and energy for an arbitrary number of links

Define the generalized-coordinate vector

$$\boldsymbol{q} = \begin{bmatrix} x & \theta_1 & \cdots & \theta_n \end{bmatrix}^{\mathsf{T}}. \tag{3}$$

For compact notation, define

$$a_{ki} = \begin{cases} L_i, & i < k, \\ \ell_k, & i = k, \\ 0, & i > k. \end{cases} \tag{4}$$

The center of mass of link k is then

$$x_k = x + \sum_{i=1}^{k} a_{ki} \sin\theta_i, \tag{5}$$

$$z_k = \sum_{i=1}^{k} a_{ki} \cos\theta_i. \tag{6}$$

Differentiation gives

$$\dot{x}_k = \dot{x} + \sum_{i=1}^{k} a_{ki} \cos\theta_i \, \dot{\theta}_i, \tag{7}$$

$$\dot{z}_k = -\sum_{i=1}^{k} a_{ki} \sin\theta_i \, \dot{\theta}_i. \tag{8}$$

The total kinetic and potential energies are

$$T = \frac{1}{2} M \dot{x}^2 + \sum_{k=1}^{n} \left[\frac{1}{2} m_k (\dot{x}_k^2 + \dot{z}_k^2) + \frac{1}{2} J_k \dot{\theta}_k^2 \right], \tag{9}$$

$$V = \sum_{k=1}^{n} m_k g z_k, \qquad \mathcal{L} = T - V. \tag{10}$$

The Euler–Lagrange equations are

$$\frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = Q_j, \qquad \boldsymbol{Q} = \begin{bmatrix} u & 0 & \cdots & 0 \end{bmatrix}^{\mathsf{T}}. \tag{11}$$

It is useful to collect the mass carried by each angular coordinate in the quantity

$$h_i = m_i \ell_i + L_i \sum_{k=i+1}^{n} m_k. \tag{12}$$

The exact configuration-dependent inertia matrix obtained from (10) has entries

$$\mathcal{M}_{00}(\boldsymbol{q}) = M + \sum_{k=1}^{n} m_k, \tag{13}$$

$$\mathcal{M}_{0i}(\boldsymbol{q}) = \mathcal{M}_{i0}(\boldsymbol{q}) = h_i \cos\theta_i, \tag{14}$$

$$\mathcal{M}_{ij}(\boldsymbol{q}) = \sum_{k=\max(i,j)}^{n} m_k a_{ki} a_{kj} \cos(\theta_i - \theta_j) + \delta_{ij} J_i. \tag{15}$$

Equations (6)–(11), or equivalently the standard form

$$\mathcal{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{e}_0 u, \tag{16}$$

are the nonlinear mechanical model. Here $\boldsymbol{e}_0 = [1, 0, \dots, 0]^\mathsf{T}$, $\boldsymbol{c}$ contains the quadratic velocity terms, and

$$\boldsymbol{g}(\boldsymbol{q}) = \begin{bmatrix} 0 & -gh_1 \sin\theta_1 & \cdots & -gh_n \sin\theta_n \end{bmatrix}^\mathsf{T}. \tag{17}$$

4 Linearization about the upright equilibrium

The upright equilibrium is

$$\theta_i = 0, \qquad \dot{x} = \dot{\theta}_i = 0, \qquad u = 0. \tag{18}$$

The cart position $x = x_\star$ may be any constant because the ideal rail is translation invariant. Around this point,

$$\sin\theta_i \simeq \theta_i, qquad \cos\theta_i \simeq 1, qquad \cos(\theta_i - \theta_j) \simeq 1. \tag{19}$$

Products such as $\theta_i \dot{\theta}_j^2$ are higher than first order and therefore disappear from the linear model. The result is

$$\mathbf{E}_n \ddot{\boldsymbol{q}} + \mathbf{D}_n \dot{\boldsymbol{q}} - \mathbf{S}_n \boldsymbol{q} = \boldsymbol{e}_0 u, \tag{20}$$

where

$$\mathbf{E}_n = \begin{bmatrix} M + \sum m_k & \boldsymbol{h}^\mathsf{T} \\ \boldsymbol{h} & \mathbf{H} \end{bmatrix}, \quad \mathbf{S}_n = \begin{bmatrix} 0 & \mathbf{0}^\mathsf{T} \\ \mathbf{0} & \operatorname{diag}(gh_1, \dots, gh_n) \end{bmatrix}, \tag{21}$$

and

$$H_{ij} = \sum_{k=\max(i,j)}^{n} m_k a_{ki} a_{kj} + \delta_{ij} J_i. \tag{22}$$

The minus sign before $\mathbf{S}_n$ is important: gravity pushes a small upright angle farther away from zero, so the equilibrium is open-loop unstable.

For future use, relative-hinge viscous damping can be represented by

$$\mathbf{D}_n = \begin{bmatrix} b_c & \mathbf{0}^\mathsf{T} \\ \mathbf{0} & \mathbf{C}_n^\mathsf{T} \operatorname{diag}(b_1, \dots, b_n) \mathbf{C}_n \end{bmatrix}, \quad \mathbf{C}_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ -1 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & 0 \\ 0 & \cdots & -1 & 1 \end{bmatrix}. \tag{23}$$

In all numerical matrices below, $\mathbf{D}_n = \mathbf{0}$ as required by (1). Dry friction is also omitted because a sign-dependent Coulomb term is nonlinear and does not have an ordinary derivative at zero velocity.

4.1 Conversion to first-order state space

Choose the state

$$\boldsymbol{z} = \begin{bmatrix}\boldsymbol{q}^\top & \dot{\boldsymbol{q}}^\top\end{bmatrix}^\top \in \mathbb{R}^{2(n+1)}. \tag{24}$$

Solving (20) for acceleration proves that

$$\boxed{\dot{\boldsymbol{z}} = \mathbf{A}_n \boldsymbol{z} + \mathbf{B}_n u, \quad \mathbf{A}_n = \begin{bmatrix}\mathbf{0} & \mathbf{I} \\ \mathbf{E}_n^{-1}\mathbf{S}_n & -\mathbf{E}_n^{-1}\mathbf{D}_n\end{bmatrix}, \quad \mathbf{B}_n = \begin{bmatrix}\mathbf{0} \\ \mathbf{E}_n^{-1}\boldsymbol{e}_0\end{bmatrix}.} \tag{25}$$

If every state is reported, then $\boldsymbol{y} = \mathbf{C}\boldsymbol{z} + \mathbf{D}u$ with $\mathbf{C} = \mathbf{I}$ and $\mathbf{D} = \mathbf{0}$.

5 Single inverted pendulum

For $\boldsymbol{q} = [x, \theta_1]^\top$, define

$$h_1 = m_1\ell_1, \qquad I_1 = J_1 + m_1\ell_1^2. \tag{26}$$

Then

$$\mathbf{E}_1 = \begin{bmatrix} M + m_1 & h_1 \\ h_1 & I_1 \end{bmatrix}, \qquad \mathbf{S}_1 = \begin{bmatrix} 0 & 0 \\ 0 & gh_1 \end{bmatrix}. \tag{27}$$

Writing $\Delta = (M + m_1)I_1 - h_1^2$, the two acceleration equations are

$$\ddot{x} = -\frac{h_1^2 g}{\Delta}\theta_1 + \frac{I_1}{\Delta}u, \tag{28}$$

$$\ddot{\theta}_1 = \frac{(M + m_1)gh_1}{\Delta}\theta_1 - \frac{h_1}{\Delta}u. \tag{29}$$

Thus, for $\boldsymbol{z} = [x, \theta_1, \dot{x}, \dot{\theta}_1]^\top$,

$$\mathbf{A}_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -h_1^2 g/\Delta & 0 & 0 \\ 0 & (M + m_1)gh_1/\Delta & 0 & 0 \end{bmatrix}, \quad \mathbf{B}_1 = \begin{bmatrix} 0 \\ 0 \\ I_1/\Delta \\ -h_1/\Delta \end{bmatrix}. \tag{30}$$

With the numerical values,

$$\mathbf{E}_1 = \begin{bmatrix} 2.5 & 0.075 \\ 0.075 & 0.0150667 \end{bmatrix}, \qquad \mathbf{S}_1 = \begin{bmatrix} 0 & 0 \\ 0 & 0.73575 \end{bmatrix}. \tag{31}$$

The numerical state-space model is

$$\mathbf{A}_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -1.72217 & 0 & 0 \\ 0 & 57.40572 & 0 & 0 \end{bmatrix}, \qquad \mathbf{B}_1 = \begin{bmatrix} 0 \\ 0 \\ 0.470221 \\ -2.340702 \end{bmatrix}. \tag{32}$$

6 Double inverted pendulum

For $\boldsymbol{q} = [x, \theta_1, \theta_2]^\mathsf{T}$,

$$h_1 = m_1\ell_1 + m_2L_1, \qquad h_2 = m_2\ell_2, \tag{33}$$

$$H_{11} = J_1 + m_1\ell_1^2 + m_2L_1^2, \qquad H_{12} = m_2L_1\ell_2, \tag{34}$$

$$H_{22} = J_2 + m_2\ell_2^2. \tag{35}$$

Therefore

$$\mathbf{E}_2 = \begin{bmatrix} M + m_1 + m_2 & h_1 & h_2 \\ h_1 & H_{11} & H_{12} \\ h_2 & H_{12} & H_{22} \end{bmatrix}, \quad \mathbf{S}_2 = \mathrm{diag}(0, gh_1, gh_2). \tag{36}$$

Substitution of the numerical values gives

$$\mathbf{E}_2 = \begin{bmatrix} 3.000000 & 0.225000 & 0.075000 \\ 0.225000 & 0.0600667 & 0.022500 \\ 0.075000 & 0.022500 & 0.0150667 \end{bmatrix}, \quad \mathbf{S}_2 = \mathrm{diag}(0, 2.20725, 0.73575). \tag{37}$$

To display the state model compactly, set $\mathbf{F}_2 = \mathbf{E}_2^{-1}\mathbf{S}_2$ and $\boldsymbol{b}_2 = \mathbf{E}_2^{-1}\boldsymbol{e}_0$. Numerically,

$$\mathbf{F}_2 = \begin{bmatrix} 0 & -4.395070 & 0.479736 \\ 0 & 102.164394 & -43.563465 \\ 0 & -130.690396 & 111.500959 \end{bmatrix}, \quad \boldsymbol{b}_2 = \begin{bmatrix} 0.466372 \\ -1.991197 \\ 0.652037 \end{bmatrix}. \tag{38}$$

For $\boldsymbol{z} = [x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]^\mathsf{T}$, the required matrices are

$$\boxed{\mathbf{A}_2 = \begin{bmatrix} \mathbf{0}_{3\times3} & \mathbf{I}_3 \\ \mathbf{F}_2 & \mathbf{0}_{3\times3} \end{bmatrix}, \qquad \mathbf{B}_2 = \begin{bmatrix} \mathbf{0}_{3\times1} \\ \boldsymbol{b}_2 \end{bmatrix}.} \tag{39}$$

7 Triple inverted pendulum

For $\boldsymbol{q} = [x, \theta_1, \theta_2, \theta_3]^\mathsf{T}$,

$$h_1 = m_1\ell_1 + (m_2 + m_3)L_1, \tag{40}$$

$$h_2 = m_2\ell_2 + m_3L_2, \tag{41}$$

$$h_3 = m_3\ell_3, \tag{42}$$

and

$$H_{11} = J_1 + m_1\ell_1^2 + (m_2 + m_3)L_1^2, \tag{43}$$

$$H_{12} = L_1(m_2\ell_2 + m_3L_2), \qquad H_{13} = m_3L_1\ell_3, \tag{44}$$

$$H_{22} = J_2 + m_2\ell_2^2 + m_3L_2^2, \qquad H_{23} = m_3L_2\ell_3, \tag{45}$$

$$H_{33} = J_3 + m_3\ell_3^2. \tag{46}$$

Consequently,

$$\mathbf{E}_3 = \begin{bmatrix} M + m_1 + m_2 + m_3 & h_1 & h_2 & h_3 \\ h_1 & H_{11} & H_{12} & H_{13} \\ h_2 & H_{12} & H_{22} & H_{23} \\ h_3 & H_{13} & H_{23} & H_{33} \end{bmatrix}, \quad \mathbf{S}_3 = \mathrm{diag}(0, gh_1, gh_2, gh_3). \tag{47}$$

The numerical matrices are

$$\mathbf{E}_3 = \begin{bmatrix} 3.500000 & 0.375000 & 0.225000 & 0.075000 \\ 0.375000 & 0.1050667 & 0.067500 & 0.022500 \\ 0.225000 & 0.067500 & 0.0600667 & 0.022500 \\ 0.075000 & 0.022500 & 0.022500 & 0.0150667 \end{bmatrix}, \quad \mathbf{S}_3 = \mathrm{diag}(0, 3.67875, 2.20725, 0.73575). \tag{48}$$

Let $\mathbf{F}_3 = \mathbf{E}_3^{-1}\mathbf{S}_3$ and $\boldsymbol{b}_3 = \mathbf{E}_3^{-1}\boldsymbol{e}_0$. Their numerical values are

$$\mathbf{F}_3 = \begin{bmatrix} 0 & -7.230245 & 1.164181 & -0.127074 \\ 0 & 161.659005 & -105.715955 & 11.539235 \\ 0 & -176.193258 & 213.836630 & -55.752848 \\ 0 & 57.696177 & -167.258544 & 115.492489 \end{bmatrix}, \quad \boldsymbol{b}_3 = \begin{bmatrix} 0.466088 \\ -1.965408 \\ 0.527435 \\ -0.172714 \end{bmatrix}. \tag{49}$$

For

$$\boldsymbol{z} = [x, \theta_1, \theta_2, \theta_3, \dot{x}, \dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3]^\mathsf{T}, \tag{50}$$

the triple-pendulum state model is

$$\boxed{\mathbf{A}_3 = \begin{bmatrix} \mathbf{0}_{4\times4} & \mathbf{I}_4 \\ \mathbf{F}_3 & \mathbf{0}_{4\times4} \end{bmatrix}, \qquad \mathbf{B}_3 = \begin{bmatrix} \mathbf{0}_{4\times1} \\ \boldsymbol{b}_3 \end{bmatrix}.} \tag{51}$$